FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VII

JEE (Main)-2020

TEST DATE: 23-08-2020

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 75 questions.
- Each subject (PCM) has 25 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only three sections: **Section-A**, **Section-B and Section-C**.

  Section-A (01 – 20, 26 – 45, 51 – 70) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 22, 46 – 47, 71 – 72) *contains 6 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

  Section-C (23 – 25, 48 – 50, 73 – 75) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

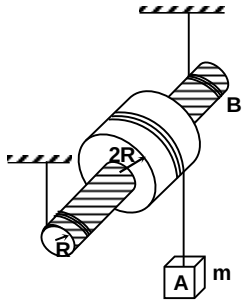
PART – I

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

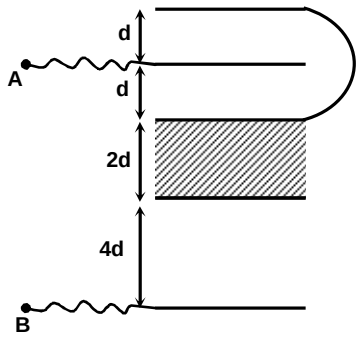
1. The volume charge density in a spherical region varies with the radial distance 'r' from its centre as $\rho = \frac{\alpha}{r}$ coulomb/m^3, where α is a constant. The electric field intensity in the spherical region varies as
   (A) constant
   (B) inversely proportional to 'r'
   (C) directly proportional to 'r'
   (D) inversely proportional to 'r^2'

2. In the arrangement as shown, if the pulley accelerates in vertical downward direction with 2 m/s^2, then the acceleration of the block is (there is no slipping between the strings and the pulley)
   (A) 2 m/s^2
   (B) 4 m/s^2
   (C) 6 m/s^2
   (D) 8 m/s^2



3. A ball is projected near the surface with speed u at an angle θ with the horizontal surface of an imaginary planet whose acceleration due to gravity on its surface is $g_p = 2g$ and wind blows with an acceleration of 2g along horizontal in the direction of throw. Find the range of the projectile. (g = acceleration due to gravity on the earth's surface)
   (A) $\frac{2u^2}{g}\sin\theta(\sin\theta+\cos\theta)$
   (B) $\frac{u^2}{2g}\cos\theta(\sin\theta+\cos\theta)$
   (C) $\frac{u^2}{g}\cos\theta(\sin\theta+\cos\theta)$
   (D) $\frac{u^2}{g}\sin\theta(\sin\theta+\cos\theta)$

4. In the arrangement shown, a dielectric slab of dielectric constant k = 2 is completely filled in the shaded region. All the plates have same area S. Find the equivalent capacitance between A and B. (Given $\frac{\varepsilon_0 S}{d} = 1\mu F$)



(A) $\frac{2}{11}\mu f$

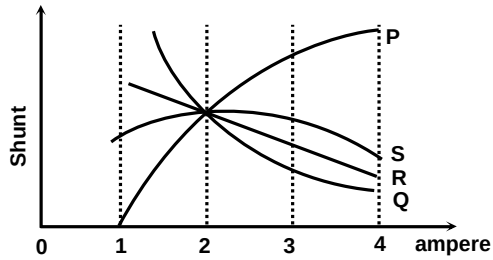
(B) $\frac{4}{11}\mu f$

(C) $\frac{7}{22}\mu f$

(D) $\frac{9}{22}\mu f$

5. The ammeter has a range of 1 ampere without shunt. The range can be varied by using different shunt resistance. The graph between the shunt resistance and the range will have the nature

(A) P
(B) Q
(C) R
(D) S



6. The Rutherford's calculation of electric potential at a distance 'r' from the centre of a spherical atom is

$V = \frac{Ze}{4\pi\varepsilon_0}\left(\frac{1}{r} - \frac{3}{2R} + \frac{r^2}{2R^3}\right)$; where R is the radius of spherical atom

The electric field intensity at a distance 'r' from the centre of spherical atom is

(A) $\frac{ze}{4\pi\varepsilon_0}\left(\frac{2r}{R^3} - \frac{1}{r^2}\right)$

(B) $\frac{ze}{4\pi\varepsilon_0}\left(\frac{1}{r^2} - \frac{2r}{3R^3}\right)$

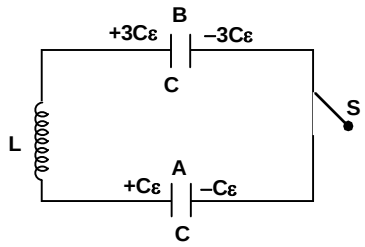
(C) $\frac{ze}{4\pi\varepsilon_0}\left(\frac{1}{r^2} + \frac{2r}{3R^3}\right)$

(D) $\frac{ze}{4\pi\varepsilon_0}\left(\frac{1}{r^2} - \frac{r}{R^3}\right)$

7. The aperture diameter of a plano-convex lens is 6 cm and thickness 3 mm at its centre. If the speed of light in the material of lens is 2×10^8 m/s, the focal length of the lens is

(A) 15 cm
(B) 20 cm
(C) 30 cm
(D) 10 cm

8. In the circuit shown, initially charge on the two capacitors A and B are $C\varepsilon$ and $3C\varepsilon$ respectively and connected through an inductor of inductance 'L' and having zero initial current. At t = 0, the switch 'S' is closed, then the maximum current through the inductor is



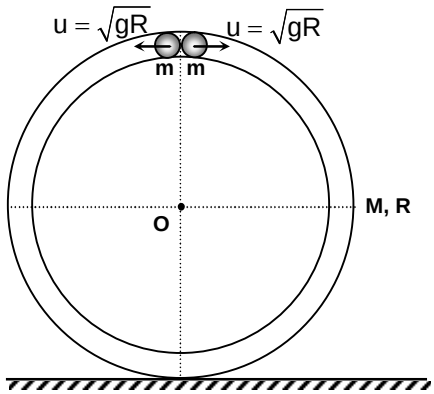
(A) $\varepsilon\sqrt{\frac{2C}{L}}$

(B) $\varepsilon\sqrt{\frac{C}{L}}$

(C) $\varepsilon\sqrt{\frac{C}{2L}}$

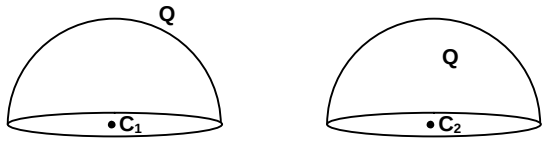
(D) $\varepsilon\sqrt{\frac{C}{3L}}$

9. A circular tube of mass 'M' and radius 'R' is placed vertically on a horizontal surface. Two identical small spheres each of mass m = 1 kg just fitting inside the tube are imparted equal and opposite initial horizontal velocity $u = \sqrt{gR}$ at the top of the tube as shown. If the tube loses contact with the horizontal surface when the radius vector of each ball makes an angle $\theta = 60°$ with the vertical, then the value of 'M' is



(A) 0.5 kg
(B) 1.5 kg
(C) 2.5 kg
(D) 4.5 kg

10. Charge Q each are distributed uniformly on a non-conducting hemispherical shell (on the surface) and in the volume of a non-conducting solid hemisphere. A point charge q is brought slowly at the centre C_1 of shell and at the centre C_2 of solid hemisphere from infinity.



Hemispherical shell Solid hemisphere

If the work done in these two processes are W_1 and W_2 then, the ratio $\left(\frac{W_1}{W_2}\right)$ is

(A) $\frac{3}{4}$

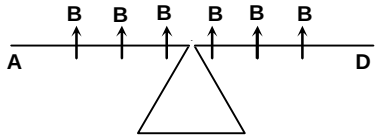
(B) $\frac{3}{2}$

(C) $\frac{4}{3}$

(D) $\frac{2}{3}$

11. The work function of a metallic surface is 3.8 eV. Photoelectrons are emitted when light of wavelength 2000 Å falls on it. The minimum potential difference required to stop the fastest photoelectron is
(A) 1.2 volts
(B) 2.4 volts
(C) 3.6 volts
(D) 4.8 volts

12. A conducting wire is bent into the shape of an equilateral triangular loop with each side of mass 'm' and length 'ℓ'. The loop is suspended in a vertical plane from a horizontal axis 'AD' as shown. A uniform vertical upward magnetic field 'B' is existing in the region. If charge 'Q' is passed through the loop almost instantaneously, then the angular velocity acquired by the loop is



(A) $\frac{3\sqrt{3}BQ}{5m}$
(B) $\frac{2\sqrt{3}BQ}{5m}$
(C) $\frac{\sqrt{3}BQ}{5m}$
(D) $\frac{BQ}{5m}$

13. A transformer is used to light a 140 watt, 24 volt lamp from 240 V AC mains. The current in the main cable is 0.7 amp. The efficiency of the transformer is
(A) 48 %
(B) 63.8%
(C) 83.3 %
(D) 90%

14. The fundamental frequency of a sonometer wire increases by 6 Hz if its tension is increased by 44% keeping the length constant. The decrease in fundamental frequency (in Hz) of the sonometer wire when the length of the wire is increased by 20%, keeping the original tension in the wire, will be
(A) 2
(B) 4
(C) 5
(D) 10

15. In Young's double slit experiment, the central bright fringe of the interference pattern produced by light of wavelength 6000 Å is found to shift to the position of 4th bright fringe after a glass plate of refractive index 1.5 is placed in front of one slit. The thickness of the glass plate is
(A) 4.8 μm
(B) 8.23 μm
(C) 14.48 μm
(D) 3.78 μm

16. If gravitational force alone prevents a spherical rotating neutron star from disintegrating and it has a mean density of magnitude $\left(\frac{9\pi}{20}\right)\times 10^{11}$ kg/m^3, then the minimum time period of star is

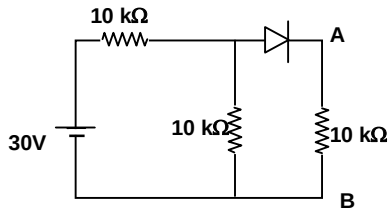
$\left(\text{Take } G=\frac{20}{3}\times 10^{-11}\ \text{Nm}^2/\text{kg}^2\right)$

(A) 1 sec
(B) 1.5 sec
(C) 3 sec
(D) 5 sec

17. A small concentration of minority charge carriers is injected into a homogenous semiconductor crystal at one point. An electric field of 10 V/cm is applied across the crystal and this moves the minority carriers a distance of 1 cm in 20 μ sec. The mobility (in cm^2/V-sec) will be
(A) 1000
(B) 2000
(C) 5000
(D) 500000

18. An immersion rod is used to heat 10 kg of water. The resistance of rod is $10^3\ \Omega$. Find the time (in minutes) for which the current 0.1 ampere be passed so that water is heated from 10°C to 20°C. (Specific heat of water = 4.2 Joule/gm°C)
(A) 350 minutes
(B) 700 minutes
(C) 1400 minutes
(D) 2100 minutes

19. In the given figure potential difference between A and B is (assume ideal diode)
(A) 0
(B) 5V
(C) 10 V
(D) 15 V

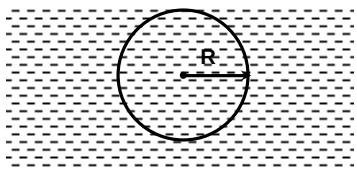


20. In a compound microscope, the objective has a focal length of 1.5 cm and the eyepiece has a focal length of 5 cm. If an object is placed at a distance of 2 cm from the objective then the magnification of the compound microscope for the adjustment for near point vision is (microscope is used by a person with near point at 25 cm).
(A) 15
(B) 18
(C) 24
(D) 30

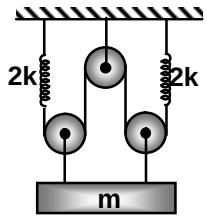
SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

21. A sphere is just immersed in a liquid. Find the ratio of hydrostatic force acting on the lower half and the upper half of the sphere. (Neglect atmospheric pressure)



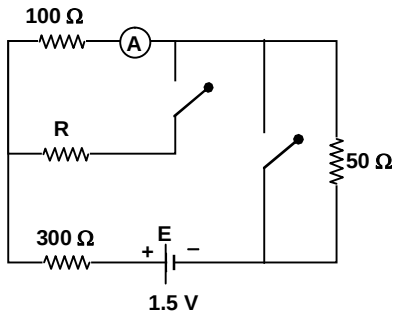
22. In the arrangement as shown if the block is slightly displaced from its equilibrium position and then released, the frequency of small oscillation of the block is $\frac{n}{2\pi}\sqrt{\frac{k}{m}}$. Find the value of 'n'. (All the springs and pulleys are ideal)



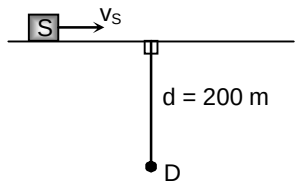
SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

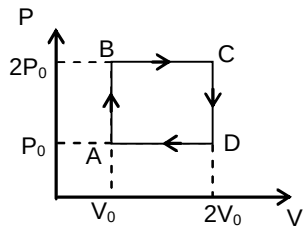
23. In the circuit shown, the reading of the ammeter (ideal) is the same with both switches open as with both switches closed. Find the value of resistance R in ohm.



24. A source emitting sound of frequency 600 Hz is moving along a straight horizontal road with a constant velocity 66 m/s. A stationary detector 'D' is located at a distance 200 m from the straight road. The velocity of sound in air is 330 m/s. Find the frequency (in Hz) of sound received by the detector when the source is closest to the detector.



25. The given figure shows the indicator diagram corresponding to n moles of an ideal monoatomic gas taken along the cyclic process ABCDA. Find the efficiency (in percentage) of the cycle.



Chemistry

PART – II

SECTION – A
(One Options Correct Type)

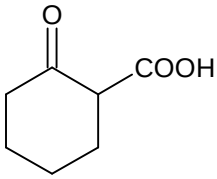
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

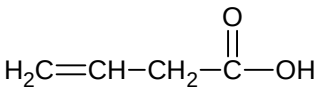
26.

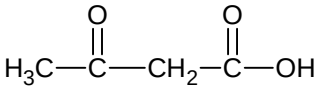
Column A	**Column B**
Name of Ore	Molecular formula
(a) Cassiterite	(i) Fe_3O_4
(b) Magnetite	(ii) SnO_2
(c) Pyrolusite	(iii) MnO_2
(d) Malachite	(iv) $CuCO_3.Cu(OH)_2$

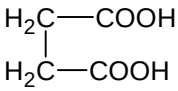
Select the correct combination:
(A) (a) (i); b (ii); (c) (iii); d(iv)
(B) (a) (iii); b (i); (c) (ii); d(iv)
(C) (a) (ii); b (i); (c) (iii); d(iv)
(D) (a) (ii); b (i); (c) (iv); d(iii)

27. Which of the following molecules will not liberate CO_2 on heating?

(A)


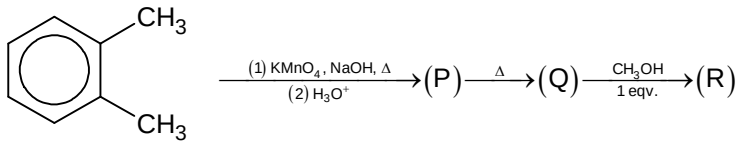
(B)


(C)


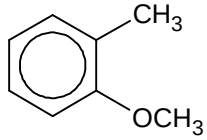
(D)


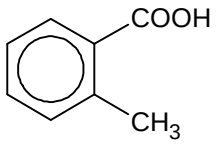
28. A quantity of 212.0 g of an aqueous solution of solute (molar mass 50.0 g) having solute mole fraction to be 0.1 is reduced to 180.0 g on evaporation and simultaneously 14.0 g of solute crystallizes out. The molality of the saturated solution left over is:
(A) 0.5
(B) 7.69
(C) 4.0
(D) 5.0

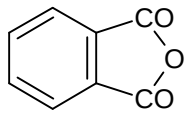
29. Which one of the following is not a common component of photochemical smog?
(A) Peroxyacyl nitrate
(B) Ozone
(C) Chlorofluoro carbons
(D) Acrolein

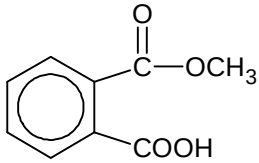
30. 

Considering the above sequence of reaction the product (R) is:

(A) 

(B) 

(C) 

(D) 

31. Which of the following statements regarding a hydrophilic sol is INCORRECT?
(A) Viscosity and surface tension are not very different from that of water.
(B) Coagulation is not very susceptible to addition of electrolytes.
(C) The charge on the sol particles depends on the pH value and it may be positive, negative or even zero.
(D) Stability towards coagulation is attributed to the electric charge on the sol particles as well as their hydration.

32. Compound 'P' (orange coloured solid) on heating gives compound 'Q' (green colour), 'R' (dry colourless gas) and water. 'R' on heating with Mg gives a compound 'S' (white solid) which on hydrolysis gives a colourless pungent smelling gas 'T'. Gas 'T' gives white fumes with conc. HCl. Gas 'R' may be
(A) N_2
(B) NH_3
(C) NO_2
(D) NO

33. $(CH_3)_2C=C(CH_3)_2 \xrightarrow{CF_3COOOH} (A) \xrightarrow[(2)\ H_2O]{(1)\ NaNH_2} (B) \xrightarrow{HNO_2} (C)$

Considering the above sequence of reaction product 'C' is
(A) $(H_3C)_2C(OH)-C(OH)(CH_3)_2$
(B) $(H_3C)_2C(OH)-C(NH_2)(CH_3)_2$
(C) $(CH_3)_3CCOCH_3$
(D) $(H_3C)_2C-C(CH_3)_2$ (epoxide, O)

34. 3.0 g of an ideal gas A, when placed in an evacuated vessel at 27°C, exerts a pressure of 1.5 atm, when a 9.0 g of another ideal gas B, non-reacting with A, is introduced in the same vessel pressure becomes 2.0 atm. What is the ratio between the average speeds of A and B at the same temperature?
(A) 1 : 3
(B) 1 : 4
(C) 4 : 1
(D) 3 : 1

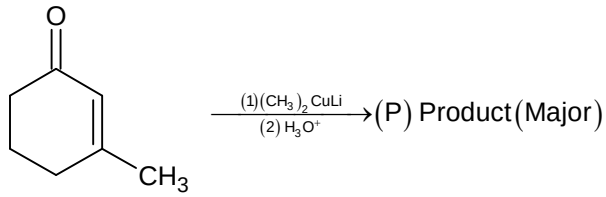
35. Which of the following statements is INCORRECT?
(A) At room temperature, ortho-hydrogen is more stable than para-hydrogen.
(B) The percentage of ortho-hydrogen increases with decrease in temperature.
(C) Ortho-hydrogen has proton of parallel spin.
(D) Ortho-hydrogen and para-hydrogen have different nuclear magnetic moment.

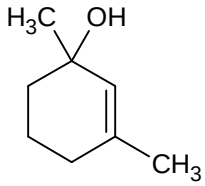
36. 0.3 M solutions each of acid HA($K_a = 1.0 \times 10^{-5}$) and HCl are neutralized separately with 0.3 M NaOH to the extent of 50%. The resulting solutions are subsequently diluted to 10 times of their current volumes. The pH values of the two solutions (final) respectively will be:
(A) 5.0 and 1.0
(B) 5.0 and 2.0
(C) 6.0 and 1.0
(D) 6.0 and 2.0

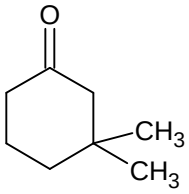
37. Which of the following statements is INCORRECT?
(A) Gypsum is added to cement to increase the setting time.
(B) Setting of cement is an exothermic process.
(C) Setting of plaster of paris is catalysed by borax.
(D) Setting of plaster of paris is catalysed by sodium chloride.

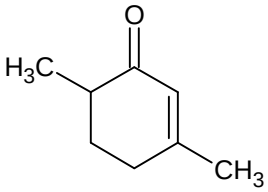
38. Which one of the following statements is INCORRECT?
(A) Aspartame is the artificial sweetner which is stable at lower temperature.
(B) Bithional is generally added to soaps as an antiseptic.
(C) Bithional is generally added as a softner to soaps.
(D) Alizarin with Al^{3+}, gives rose red colour.

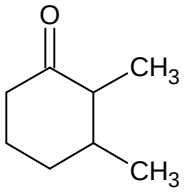
39. Which of the following alloy does not contain copper?
(A) Bell metal
(B) Type metal
(C) Monel metal
(D) Gun metal

40. 

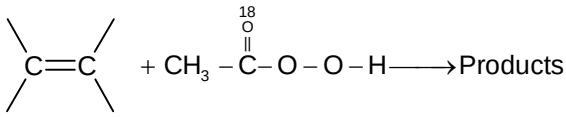
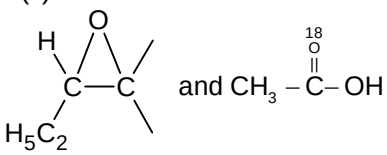
The product 'P' is:
(A) 

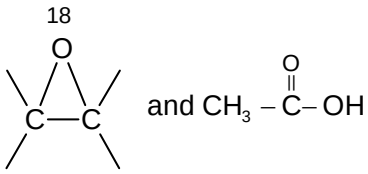
(B) 

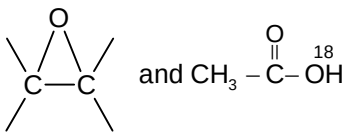
(C) 

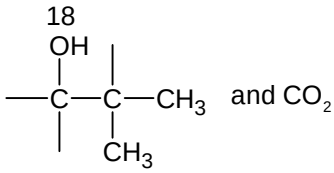
(D) 

41. Which of the following molecules showing bridge bonding is planar ?
(A) B_2H_6
(B) I_2Cl_6
(C) Al_2Cl_6
(D) $Al_2(CH_3)_6$

42. 
The product(s) are
(A) 

(B) 

(C) 

(D) 

43. Which of the following statements is CORRECT?
I. Passing of H_2S gas through nitric acid produces plastic sulphur.
II. Passing of H_2S gas through nitric acid produces colloidal sulphur.
III. Passing of H_2S gas through SO_2 produces colloidal sulphur.
(A) Only I
(B) I and III
(C) II and III
(D) I, II and III

44. Which of the following tests can be used to distinguish between glucose and fructose?
(A) Tollen's test
(B) Furfural test
(C) Fehling's test
(D) Molish's test

45. Which of the following is not an intensive property?
(A) Dielectric constant
(B) Heat capacity
(C) Viscosity
(D) Standard electrode potential

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

46. Two isotopes 'X' and 'Y' of atomic mass 10 and 20 respectively are mixed in equal amounts by mass. After 30 days, their mass ratio is found to be 4 : 1. If isotope 'X' has half-life of 10 days. What is the half-life of 'Y' (in days)?

47. How many of the following are copolymers
Polyurethrane ; Polymethylacrylate ; Glyptal ; Nylon-6,6 ; Teflon ; Dacron ; Bakelite ; BUNA-S ; Polyacrylonitrile.

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

48. An organic compound was analysed by Duma's method, 0.40 g of compound on combustion gave 50 ml of nitrogen at 27°C and 756 mm of pressure. Calculate percentage of nitrogen in the compound.

49. The orbital angular momentum of an electron is $\sqrt{3}\frac{h}{\pi}$, if the minimum permissible value of total energy (eV) of this electron in unknown Bohr's orbit of Li^{2+} is – x eV, then the value of 'x' is :

50. $COCl_2(g)$ dissociates in the following manner
$$COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$$
When $COCl_2(g)$ is heated to 750 K at 1.0 atm, density of the gas mixture was found to be 1.162 g/L. What is the value of K_P? [Take value of R = 0.082, and (assume the reaction is started taking 1 mole of $COCl_2$ initially)]

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

51. If 1 – i be one of the roots of $ax^2 - bx + c = 0$ where a, b, c are the sides of a triangle ABC, then which option is incorrect?
(A) Triangle is isosceles
(B) Greatest angle of the triangle is $\cos^{-1}\left(\frac{1}{4}\right)$
(C) Smallest angle of the triangle is $\cos^{-1}\left(\frac{7}{8}\right)$
(D) a, 2b, 3c are in Arithmetic Progression

52. The sum of all terms in 11th bracket of the series
$(1) + (2 + 2^2) + (2^3 + 2^4 + 2^5) + (2^6 + 2^7 + 2^8 + 2^9) + ...$ is $2^p - 2^q$, then $\frac{2p+q}{2p-q}$ is
(A) 17 : 13
(B) 17 : 7
(C) 17 : 6
(D) none of these

53. If the number of quadrilaterals be just double of the number of triangles that can be formed by joining the vertices of a polygon. If the number of sides of polygon is n, then
(A) n is a multiple of 5
(B) n is a multiple of 3
(C) n is a multiple of 7
(D) n is a multiple of 11

54. If $A = \begin{bmatrix} \sqrt[3]{8} & 0 & 0 \\ 0 & \sqrt[3]{2} & 0 \\ 0 & 0 & \frac{1}{\sqrt[3]{4}} \end{bmatrix}$ and $\left|\text{Adj}(\text{Adj } A)\right| = \sqrt[3]{\lambda^{\lambda}}$, then $2\lambda - 1$ is
(A) 5
(B) 7
(C) 9
(D) 11

55. If p be the probability of arranging the word 'BANANA' in which A and B occur at the ends, then 3p is not equal to
(A) perpendicular distance of $3x - 4y - 1 = 0$ from origin
(B) eccentricity of conic $\frac{x^2}{25} + \frac{y^2}{24} = 1$
(C) $\lim_{x\to\infty} \frac{[x] - 5\ln x}{5(x - \{x\})}$
(where [.] & {.} denotes the greatest integer function and fractional part respectively)
(D) distance between $3x - 4y + 5 = 0$ and $3x - 4y + 10 = 0$

56. If $\theta \in [-2\pi, 2\pi]$, then number of solution of
$\sqrt{2\cos^2\theta - 4\cos\theta + 6} + \sqrt{19 - 6\cos\theta + 3\cos^2\theta} = 4 + \sin^2\theta + 2\cos\theta$
(A) 2
(B) 3
(C) 4
(D) 5

57. If $\operatorname{cosec}\theta = \cot\theta + 1$, then one of the roots of
$(a + 2b + 5c)x^2 - (3a - 5b + 4c)x + (2a - 7b - c) = 0$
(A) $\sin\theta$
(B) $\cos\theta$
(C) $\tan\theta$
(D) $\cot\theta$

58. If $\cos^{-1}\left(\frac{x}{2}\right) + \cos^{-1}\left(\frac{y}{3}\right) = \theta$ and $P = 9x^2 + 4y^2 - 12xy\cos\theta$, then which of the following is INCORRECT?
(A) $\frac{1}{18}\int_0^{\pi} P\,d\theta = \pi$
(B) $\int_0^{\pi/2} \frac{\sqrt{P}\,d\theta}{\sqrt{P} + \sqrt{36 - P}} = \frac{\pi}{4}$
(C) Minimum value of P is –36
(D) Maximum value of P is 36

59. If I be the incentre of ΔABC, P and Q are feet of perpendicular from A to BI, CI respectively. Then $\frac{AQ}{CI} + \frac{AP}{BI} = ?$
(A) $\cot\left(\frac{A+B}{2}\right)$
(B) $\cot\left(\frac{B+C}{2}\right)$
(C) $\cot\left(\frac{A+C}{2}\right)$
(D) $\tan\left(\frac{B+C}{2}\right)$

60. If f(x) is the derivative of $\tan^{-1}\left(\frac{x}{\sqrt{1+x^2}-1}\right)$, then $\frac{1}{f(x)} + \frac{1}{f(x^2)}$ is
(A) polynomial of degree 4
(B) divisible by $(x^2 + 1)$
(C) divisible by $(2x^2 + 1)$
(D) zero

61. If tangent at any point $x^4 + y^4 = c$ $(c > 0)$ cuts intercept a and b respectively from both axes such that $a^{-\alpha} + b^{-2\beta} = c^{-\gamma}$, then
(A) α, β, γ are in Arithmetic Progression
(B) α, β, γ are in Geometric Progression
(C) α, β, γ are in Harmonic Progression
(D) $\beta, \alpha, 2\gamma$ are in Arithmetic Progression

62. If $\int \frac{\sum_{r=1}^{1011} x^{-(2r-1)}}{\sum_{r=0}^{1010} x^{2r}} dx = c - \frac{1}{k \cdot x^k}$, then the value of k is

(A) –2020
(B) 2020
(C) 2022
(D) –2022

63. $I_1 = \int_1^3 (x^2 + x + 3) f(x^3 - 2x^2 - 5x + 2020) dx$, $I_2 = \int_1^3 (4x^2 - 3x - 2) f(x^3 - 2x^2 - 5x + 2020) dx$, then $2I_1 : 3I_2$

(A) $\in (1, 2)$
(B) $\in (0, 1)$
(C) $\in (-2, 0)$
(D) $\in (2, 3)$

64. The area bounded by the curve $y = e^{-x}$, $y = e^x$ and directrix of $y^2 - 4x - 4y + 12 = 0$ is

(A) $\sum_{r=1}^{\infty} \frac{1}{(2r)!}$
(B) $2\sum_{r=1}^{\infty} \frac{1}{(2r)!}$
(C) $\sum_{r=0}^{\infty} \frac{1}{(2r)!}$
(D) none of these

65. If $f''(x) = g''(x)$ and $f'(2019) = g'(2019)$ and $f(2018) = 3$ and $g(2018) = 2$, then $f(2020) - g(2020) = ?$

(A) 2020
(B) 2018
(C) 2019
(D) none of these

66. The curve represented by $2x + \sqrt{9 - 4xy + y^2 + 4x^2} = y + 1$ is

(A) straight line whose slope is 2
(B) parabola whose latus rectum is 2
(C) ellipse with eccentricity $\frac{1}{3}$
(D) hyperbola whose eccentricity is $\sqrt{2}$

67. The statement $\sim p \vee \sim q$ is equivalent to

(A) $\sim q \vee p$
(B) $-(p \wedge q)$
(C) $q \vee p$
(D) none of these

68. If the eccentricities of $\frac{x^2}{7}-\frac{y^2}{b^2}=1$ and $\frac{x^2}{16}+\frac{y^2}{9}=1$ are reciprocal to each other, then area bounded by the curve $\frac{x^2}{25}+\frac{y^2}{b^2}=1$ is

(A) 10π
(B) 15π
(C) 20π
(D) 25π

69. If the line $\frac{x-6}{3}=\frac{y-2}{5}=\frac{z+3}{2}$ meets the curve $xy = a^2$, $z = 1$, then the number of real value of a is

(A) 0
(B) 1
(C) 2
(D) more than 2

70. If $\vec{a}$ and $\vec{b}$ are two vectors satisfying $\vec{a}+\vec{b}=\vec{p}$ and $2\vec{a}+\vec{b}=\vec{q}$ where $\vec{p}=2\hat{i}-3\hat{j}$ & $\vec{q}=-\hat{i}-\hat{j}$. Angle between $\vec{a}$ & $\vec{b}$ is θ such that $\cos\theta+\sqrt{\frac{\alpha}{\alpha+1}}=0$, then the value of α is

(A) 20
(B) 24
(C) 25
(D) 26

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

71. If H(α, β), G(α_1, β_1) and C(α_2, β_2) be the orthocentre, centroid and circumcentre respectively of the triangle formed by the lines connected by $2xy = x^2y + xy^2$. Then the value of $5\alpha - 4\beta + 3\alpha_1 + 3\beta_1 - 2\alpha_2 - \beta_2$ is

72. $\underbrace{272727.....27}_{34-\text{digits}} = A\left(10^{34}-1\right)$, then the value of 11A is

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

73. Standard deviations of first 10 natural numbers is

74. Consider a point 'P' with abscissa $n(n \in I^+)$ on the curve $y^2 - x^2 = 1$. If d_n represents the shortest distance from the point P to the lie y = x, then $\lim\limits_{n\to\infty}\left(4n\cdot d_n\right)$ has the value

75. If N be the number of positive integral ordered pair of (a, b) such that 6, a, b are in Harmonic Progression, then the value of $\frac{8N}{3}$ is